

3.1.6 Chemical equilibria, Le Chatelier's principle and K_c

In contrast with kinetics, which is a study of how quickly reactions occur, a study of equilibria indicates how far reactions will go. Le Chatelier's principle can be used to predict the effects of changes in temperature, pressure and concentration on the yield of a reversible reaction. This has important consequences for many industrial processes. The further study of the equilibrium constant, K_c , considers how the mathematical expression for the equilibrium constant enables us to calculate how an equilibrium yield will be influenced by the concentration of reactants and products.

3.1.6.1 Chemical equilibria and Le Chatelier's principle

Content	Opportunities for skills development
Many chemical reactions are reversible. In a reversible reaction at equilibrium: • forward and reverse reactions proceed at equal rates • the concentrations of reactants and products remain constant Le Chatelier's principle. Le Chatelier's principle can be used to predict the effects of changes in temperature, pressure and concentration on the position of equilibrium in homogeneous reactions. A catalyst does not affect the position of equilibrium. Students should be able to: • use Le Chatelier's principle to predict qualitatively the effect of changes in temperature, pressure and concentration on the position of equilibrium • explain why, for a reversible reaction used in an industrial process, a compromise temperature and pressure may be used.	PS 1.1 Students could carry out test-tube equilibrium shifts to show the effect of concentration and temperature (eg $\text{Cu}(\text{H}_2\text{O})_6^{2+}$ with concentrated HCl).

3.1.6.2 Equilibrium constant K_c for homogeneous systems

Content	Opportunities for skills development
The equilibrium constant K_c is deduced from the equation for a reversible reaction. The concentration, in mol dm^{-3}, of a species X involved in the expression for K_c is represented by [X] The value of the equilibrium constant is not affected either by changes in concentration or addition of a catalyst. Students should be able to: • construct an expression for K_c for a homogeneous system in equilibrium • calculate a value for K_c from the equilibrium concentrations for a homogeneous system at constant temperature • perform calculations involving K_c • predict the qualitative effects of changes of temperature on the value of K_c	MS 0.3 Students estimate the effect of changing experimental parameters on a measurable value, eg how the value of K_c would change with temperature, given different specified conditions. MS 1.1 Students report calculations to an appropriate number of significant figures, given raw data quoted to varying numbers of significant figures. Students understand that calculated results can only be reported to the limits of the least accurate measurement. MS 2.2 and 2.3 Students calculate the concentration of a reagent at equilibrium. Students calculate the value of an equilibrium constant K_c PS 2.3 Students could determine the equilibrium constant, K_c, for the reaction of ethanol with ethanoic acid in the presence of a strong acid catalyst to ethyl ethanoate.